

Online Retail Sales Analysis and Interactive Dashboard Report

Dataset Chosen

For this project, we used the **Online Retail dataset**. This dataset contains real-world e-commerce transaction data from an online retail business. It includes information such as invoice number, stock code, product description, quantity sold, invoice date, unit price, customer ID, and country.

We chose this dataset because it is suitable for business and sales analysis. It has more than 100 rows, more than 4 columns, and includes a time-based column called **InvoiceDate**, which makes it useful for trend analysis and dashboard development.

Data Cleaning and Preparation

The dataset was loaded using Python and Pandas. First, we inspected the structure of the dataset using `df.head()`, `df.shape`, `df.info()`, and `df.isnull().sum()`.

The dataset had missing values in the **Description** and **CustomerID** columns. Rows with missing product descriptions were removed because product information was important for the analysis. The missing **CustomerID** values were kept because customer ID was not required for the main sales, country, product, and time-based analysis.

```
df = df.dropna(subset=["Description"])
```

```
df["Revenue"] = df["Quantity"] * df["UnitPrice"]
```

A new column called **Revenue** was created using:

```
Revenue = Quantity × UnitPrice
```

This became the main column used for sales analysis, visualizations, and dashboard KPI cards.

The `InvoiceDate` column was converted to datetime format using `pd.to_datetime()`. This allowed us to create new time-based columns such as `Year`, `Month`, `MonthName`, `Day`, and `Date`.

We also checked for duplicate rows. The dataset had **5,225 duplicate records**, which were removed to avoid double-counting sales and revenue.

Exploratory Data Analysis

During EDA, we analyzed revenue, quantity, products, countries, and dates. We calculated basic statistics such as mean, median, minimum, and maximum revenue values.

We also performed group analysis to find:

- Top products by revenue
- Revenue by country
- Daily revenue trends
- Highest revenue dates
- Revenue distribution across countries

Several visualizations were created, including a bar chart, line chart, pie chart, and box plot.

The bar chart showed the top 10 products by revenue. The line chart showed daily revenue trends over time. The pie chart showed the revenue share by country, especially for the top-selling product. The box plot showed how revenue values were distributed across top countries and helped identify outliers.

Key Insights

The analysis showed that some products generated much higher revenue than others. One of the strongest products was **REGENCY CAKESTAND 3 TIER**, which appeared as a top revenue-generating product.

The United Kingdom generated the highest overall revenue compared to other countries. This suggests that the retailer has a strong customer base in the UK.

The daily revenue trend showed that revenue was not constant across time. Some days had much higher sales than others, showing peak business periods.

The box plot showed that there were outliers in the revenue data. This means that most transactions were relatively small, but a few transactions had very high revenue values.

Interactive Dashboard

An interactive dashboard was built using **Dash and Plotly**. The dashboard includes filters for:

- Month
- Country
- Product

The dashboard also includes KPI cards showing:

- Total Revenue
- Total Orders
- Average Transaction Value

The dashboard visualizations include:

- Top 10 Products by Revenue
- Daily Revenue Trend
- Revenue Share by Country
- Revenue Distribution by Country

The charts are dynamic, meaning they update when users change the filters. This makes the dashboard useful for exploring the data from different business perspectives.

Challenges Faced

One challenge was handling missing values, especially in the `CustomerID` column. Since many customer IDs were missing, we decided not to remove those rows because they still contained useful sales information.

Another challenge was dealing with duplicate rows. Removing duplicates was important because duplicate transactions could increase revenue totals incorrectly.

Long product names also created layout issues in charts, so horizontal bar charts were used to make the product names easier to read.

Another challenge was running Dash inside Jupyter/Colab. After testing different methods, the dashboard worked using `app.run(jupyter_mode="inline", debug=True)`.

What We Learned

Through this project, we learned how to take raw data and turn it into useful business insights. We practiced data cleaning, handling missing values, removing duplicates, fixing data types, and creating new columns.

We also learned how to perform exploratory data analysis using Pandas and create visualizations using Matplotlib and Plotly. Finally, we learned how to build an interactive dashboard using Dash with filters, KPI cards, and dynamic charts.

This project helped us understand how data analysts work with real-world data to support business decisions.